INCH–POUND
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MILITARY SPECIFICATION

DISK, COLOR STANDARDS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers disks for use in comparators conforming to MIL-C-538.

1.2 Classification. The disks shall be of the following types and classes, as specified (see 6.2):

Type I - pH.

Class 1 – Range 3.0 to 4.6.
Class 2 – Range 4.4 to 6.0.
Class 3 – Range 6.0 to 7.6.
Class 4 – Range 7.6 to 9.2.
Class 5 – Range 6.8 to 8.4.
Class 6 – Range 8.6 to 10.2.
Class 7 – Range 3.0 to 7.0.
Class 8 – Range 3.0 to 11.0.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6630

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Type II - Chlorine, ppm.

Class 1 – Range 0.1 to 5.0.
Class 2 – Range 3.0 to 10.0.
Class 3 – Range 0.1 to 2.0.
Class 4 – Range 0.1 to 10.0.
Class 5 – Range 0.1 to 10.0.
Class 6 – Range 0 to 250.
Class 7 – Range 0.1 to 10.0.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

L-P-378	- Plastic Sheet and Strip, Thin Gauge, Polyolefin.
PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-676	- Boxes, Setup.

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-I-536	- Indicator Solutions, pH Determination (for Water Supply Testing).
MIL-C-538	- Comparator, Color.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of U.S. Military Property.

(Unless otherwise indicated, copies of Federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

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2.1.2 Other Government documents and drawings. The following other Government documents and drawings form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those in effect on the date of the solicitation.

DRAWINGS

ME

TA13200E7430 - Disks, Color Standards.

(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained from the USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-FSQ, Fort Belvoir, VA 22060-5606.)

DEPARTMENT OF COMMERCE, NATIONAL BUREAU OF STANDARDS

Research Paper 2619 - pH Values of the Clark and Lubs Buffer Solutions at 25 °C.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

(Copies of specifications, standards, drawings, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN PUBLIC HEALTH ASSOCIATION, INC. (APHA)

Standard Methods for the Examination of Water and Wastewater.

(Application for copies should be addressed to the American Public Health Association, Inc., 1015 Fifteenth Street, NW, Washington, D.C. 20005.)

AMERICAN WATER WORKS ASSOCIATION, INC. (AWWA)

Journal, Volume 35, No. 8, August 1943 - Colormetric Determination of Chlorine Residuals up to 10 PPM with Orthotolidine.

(Application for copies should be addressed to the American Water Works Association, Inc., 6666 West Quincy Avenue, Denver, CO 80235.)

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NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc., ATTN: Traffic Order Section, 2200 Mill Road, Alexandria, VA 22314.)

UNIFORM CLASSIFICATION COMMITTEE AGENT

Uniform Freight Classification Rules

(Application for copies should be addressed to Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 S. Riverside Plaza, Chicago, IL 60606.)

(Nongovernment standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The disks shall be in accordance with TA13200E7430 and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the disks. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Materials. Materials shall be as specified herein and on the drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

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3.3.1 Material deterioration prevention and control. The disks shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operation and storage environment to which the disks may be exposed.

3.3.2 Identification of material and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and subcomponents, and shall make information available, upon request, to the contracting officer or designated representative.

3.3.3 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the disks may be newly fabricated from recovered materials to the maximum extent practicable, provided the disks produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the disks.

3.3.4 Circular inserts. Inserts identifying the value of glass standards shall be plastic material, black printing on white or tan background.

3.4 Construction.

3.4.1 Circular inserts. The identifying circular inserts shall conform to figure 1, item B. Inserts identifying the value of the glass color standards shall be permanently fastened at a location 140 degrees counterclockwise from the glass color standards.

3.4.2 Glass color standards and filters. The glass color standards shall conform to figure 1, item A. The glass filters, when required, shall conform to figure 1, item C, and shall be located in the center of the disk. Glass color standards and filters shall be free of cracks, scratches, bubbles, blisters, inclusions, nonuniformity of color, and other defects that will adversely affect their use. When tested as specified in 4.5.2.5, glass color standards shall withstand without damage the thermal shock incurred when removed from solid carbon dioxide (dry ice) and immediately immersed in boiling water.

3.5 Performance.

3.5.1 Characteristics and accuracy.

3.5.1.1 Type I (pH) disk. The characteristics and accuracy of type I disks shall conform to table I, and, unless otherwise specified (see 6.2), shall be standardized for use with the indicator solutions specified, using viewing tubes having a 26 millimeters (mm) viewing depth. The coloring of the glass color standards shall be such that the same reading shall be obtained when used with diffused daylight, incandescent light, or white fluorescent light.

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3.5.1.2 Type II (chlorine) disks. When tested as specified in 4.5.2.4 or 4.5.2.7, as applicable, the characteristics of type II disks shall conform to table II and the accuracy of classes 1, 2, 3, 6, and 7 disks shall conform to table III. Unless otherwise specified (see 6.2), the disks shall be standardized for use with acid orthotolidine indicator solution using viewing tubes having a 26 mm viewing depth. The class 4 and 5 disks shall match the master color standard disk (see 3.8). The coloring of the glass color standard shall be such that the same reading will be obtained when used with diffused daylight, incandescent light, and white fluorescent light.

TABLE I. Characteristics and accuracy of type I disks.

Class	Range	Interval	Tolerance	Indicator solutions
1	3.0 to 4.6	0.2	±0.04	Bromophenol blue conforming to CAS 62625-28-9
2	4.4 to 6.0	0.2	±0.04	Bromocresol green- bromocresol purple conforming to MIL-I-536, type I
3	6.0 to 7.6	0.2	±0.04	Thymol Blue = 3,3-Dibromothymolsulfonaphthalein
4	7.6 to 9.2	0.2	±0.04	Cresol red-thymol blue conforming to MIL-I-536, type II
5	6.8 to 8.4	0.2	±0.04	Phenol red D
6	8.6 to 10.0	0.2	±0.04	Phenolphthalein D, commercial grade
7	3.0 to 7.0	0.5	±0.04	Wide range indicator solution
8	3.0 to 11.0	1.0	*	Wide range indicator solution

* Shall match master color standard disk (see 3.8) or buffered indicator solution specified in 4.5.2.1. If procedures specified in 4.5.2.1 are used, tolerance on each value shall be ±0.1 pH.

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TABLE II. Characteristics of type II disks.

Class	Range	Residual chlorine values in ppm
1	0.1 to 5.0	0.1, 0.3, 0.5, 0.75, 1.0, 1.5, 2.0, 3.0, and 5.0
2	3.0 to 10.0	3.0, 3.5, 4.0, 4.5, 5.0, 6.0, 7.5, 8.0, and 10.0
3	0.1 to 2.0	0.1, 0.15, 0.2, 0.35, 0.5, 0.75, 1.0, 1.4, and 2.0
4	0.1 to 10.0	0.1, 0.15, 0.25, 0.5, 1.0, 1.5, 2.5, 5.0, and 10.0
*5	0.1 to 10.0	
6	0.5 to 250	0.1, 0.4, 0.7, 1.0, 1.5, 2.0, 3.0, 5.0, and 10.0
**7	0.1 to 10.0	0.5, 15, 25, 50, 75, 100, 150, 200, and 250
		0.1, 0.4, 0.7, 1.0, 1.5, 2.0, 3.0, 5.0, and 10.0

* Shall match master color standard.
** Shall be a DPD disk which determines free chlorine.

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TABLE III. Accuracy of type II, classes 1, 2, 3, 4, 6, and 7, disks in ppm.

Class 1	Class 2	Class 3	Class 4	Class 6	Class 7
0.1 + .05, - .01	3.0 +/- .12	0.1 +/- .01	0.1 +/- .01	0.5, - .01, + 4	Tolerance 10 ppm shall be +/- 0.5, tolerance on all other ppm values shall be +/- 10 percent of the value
0.3 +/- .05	3.5 +/- .12	0.15 +/- .01	0.15 +/- .01	15 + 2.5, - 4	
0.5 +/- .06, - .05	4.0 +/- .12	0.20 + .04, - .01	0.25 + .04, - .01	25 + 6, - 2.5	
0.75 +/- .06	4.5 +/- .12	0.35 +/- .04	0.5 + .06, - .04	50 +/- 6	
1.0 + .12, - .06	5.0 + .25, - .12	0.5 + .06, - .04	1.0 + .10, - .07	75 +/- 6	
1.5 +/- .12	6.0 + .038, - .25	0.75 +/- .07	1.5 + .15, - .10	100 + 12.5, - 6	
2.0 + .25, - .12	7.5 + .12, - .38	1.0 + .10, - .07	2.5 + .50, - .25	150 +/ - 12.5	
3.0 + .50, - .25	8.0 + .50, - .12	1.4 + .15, - .10	5.0 +/- .50	200 +/- 12.5	
5.0 +/- .50	10.0 +/- .50	2.0 +/- .15	10.0 +/- .50	250 +/- 12.5	

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3.6 Product identification.

3.6.1 Specification part number. Definitive specification part numbers for items described in this specification shall be formatted as shown in 6.7 and shall be used in product identification (see 3.6.2.1.1 and 3.6.2.1.2).

3.6.2 Identification marking. The disks shall be identified in accordance with MIL-STD-130.

3.6.2.1 Special marking. Identification marking shall be applied to a plastic insert conforming to figure 1, item D, black printing on white or tan background.

3.6.2.1.1 Disk except type II, class 6. All disks except type II, class 6 shall be identified as follows:

(Name of indicator for which disk is used.)
Use 26 mm tubes.
15 milliliter (ml) sample.
(Amount of indicator or reagent used.)
(Name or registered trademark of the manufacturer.)
(Specification part number.)

3.6.2.1.2 Disk type II, class 6. The type II, class 6 disk shall be identified as follows:

(Name of indicator for which disk is to be used.)
Use 26 mm tubes.
Fill tube with 15 ml of reagent and then add 0.75 ml of water sample.
(Name or registered trademark of the manufacturer.)
(Specification part number.)

3.7 Items to be furnished with type II, class 6 disk. Each type II, class 6 disk shall be furnished with a calibrated rubber-nipple dropper graduated and numbered for 0.75 ml.

3.8 Government-loaned property. Unless otherwise specified (see 6.2), the following property in the quantities indicated will be loaned by the Government (see 6.4):

Item	Description	Identification	Quantity
1.	Comparator, color (Disk evaluator)	6630-00-087-1838	1
2.	Disk, color standard Type I, class 8 Master)	6630-00-926-1288	1
3.	Disk, color standard Type II, class 5 Master)	6630-00-926-1289	1

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3.9 Workmanship. Disks shall be free of warp, twist, bubbles, cracks, chipped edges, surface blisters, uneven surfaces, scratches, dents, heat marks, fins, burrs, nonuniformity of color, projection ridges at dividing line of mold, and unsightly finish caused by chipping, filing, or grinding of defects. Buffing or polishing may be employed if necessary.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 First article examination. The first article shall be examined as specified in 4.5.1. The presence of one or more defects shall be cause for rejection.

4.3.2 First article tests. The first article shall be tested as specified in 4.5.2, in the order shown. Failure of any test shall be cause for rejection.

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4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and thermal shock shall be in accordance with MIL-STD-105, table I and columns 1 and 2 of table II-A. Inspection levels shall be level II for examination and S-2 for thermal shock.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. Acceptance number shall be zero for both major and minor defects. No defects shall be allowed.

4.4.3 Tests.

4.4.3.1 Individual. The colored glass standards of each disk shall be subjected to the tests marked "X" in column 2 of table IV. Failure of any test shall be cause for rejection.

4.4.3.2 Samples. Samples selected in accordance with 4.4.1 shall be subjected to the tests marked "X" in column 3 of table IV. Quality conformance testing shall be performed in accordance with 4.5.2.1, 4.5.2.3 and 4.5.2.7. Sampling shall be in accordance with MIL-STD-105, table I and columns 1 and 2 of table II-A. Inspection level shall be S-2 and acceptance number shall be zero. No defects shall be allowed.

4.5 Inspection.

4.5.1 Examination. The disks shall be examined for the following defects:

Number	Defects	Requirement paragraph
Major		
101.	Material not as specified.	3.3
102.	Materials not resistant to corrosion and deterioration or treated to be made resistant to corrosion and deterioration.	3.3.1
103.	Material and material finish or treatment information not available.	3.3.2
104.	Used components are incorporated in the disks.	3.3.3
105.	Circular inserts material not as specified.	3.3.4
106.	Circular inserts identifying the value of glass color standards are not permanently located as specified.	3.4.1
107.	The glass filters are not located in the center of the disk as specified.	3.4.2
108.	The glass color standard does not withstand thermal shock when tested as specified.	3.4.2
109.	The glass color standards and filters are not free of cracks, scratches, bubbles, blisters, and nonuniformity of colors as specified.	3.4.2
110.	Workmanship not as specified.	3.9

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Number	Defects	Requirement paragraph
Minor		
201.	Marking not as specified.	3.6.2, 3.6.2.1
202.	Calibrated rubber-nipple dropping pipette not provided as specified.	3.7

4.5.2 Tests.

4.5.2.1 Type I, classes 1, 2, 3, 4, 5, 6, and 7 disks. Disks shall be visually compared with buffer solutions and indicator solutions in a comparator using sample tubes having a 26 mm viewing depth. Buffer solutions shall be prepared in accordance with National Bureau of Standards Research Paper 2619. Indicator solutions shall conform to table I. Nonconformance to table I shall constitute failure of this test.

4.5.2.2 Type II, class 1, 2, 3, or 4 disks. Disks shall be visually compared with a series of liquid chlorine standards in a comparator using sample tubes having a 26 mm viewing depth. The liquid chlorine standards shall consist of sodium hypochlorite solutions treated with orthotolidine solution as specified in the procedure outlined in "Colormetric Determination of Chlorine Residuals up to 10 ppm with Orthotolidine" of the Journal of American Water Works Association. The liquid chlorine solutions shall be standardized by titration as described in the above article and shall be matched visually with the Modified Scott permanent chlorine standards immediately before they are matched with the color disk. Preparation of the Examination of Water, Sewage, and Industrial Wastes", published by the American Public Health Association. Nonconformance to tables II or III shall constitute failure of this test.

4.5.2.3 Type I, class 8; type II, class 5 disks. Disks shall be visually compared with the applicable master color disk in a color comparator (see 3.8). For acceptance, identical match of all color standards of the offered disks with the applicable color standards of the master color disks as required. If color match of standards on offered disks and master disks are not identical, test method 4.5.2.1 may be used as an alternate. Reagents for testing pH 11 shall be prepared by adapting the procedure described in National Bureau of Standards Research Paper 2619.

4.5.2.4 Type II, class 6 disk. Disks shall be visually compared with a series of liquid chlorine standards in a comparator using sample tubes having a 26 mm viewing depth. The sample tubes shall be filled with 15 ml of orthotolidine solution and to this is added 0.75 ml of chlorine standard. The liquid chlorine standards shall be prepared using the method described in Standard Methods for the Examination of Water and Wastewater, American Public Health Association, Part I, Chlorine (Residual), B, Orthotolidine Method, Section 3. As an alternate, a standard color disk may be used for comparison, provided the disk has been previously standardized as stated above.

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TABLE IV. Test schedule.

First Article 1	Quality Conformance		Test 4	Test Paragraph 5	Requirement Paragraph 6
	Individual 2	Sample 3			
			Disk, type I (pH):		
X	X	X	Class 1	4.5.2.1, 4.5.2.6	3.5.1.1, table III
X	X	X	Class 2	4.5.2.1, 4.5.2.6	3.5.1.1, table III
X	X	X	Class 3	4.5.2.1, 4.5.2.6	3.5.1.1, table III
X	X	X	Class 4	4.5.2.1, 4.5.2.6	3.5.1.1, table III
X	X	X	Class 5	4.5.2.1, 4.5.2.6	3.5.1.1, table III
X	X	X	Class 6	4.5.2.1, 4.5.2.6	3.5.1.1, table III
	X	X	Class 7	4.5.2.1, 4.5.2.6	3.5.1.1, table III
X	X	X	Class 8	4.5.2.3, 4.5.2.6	3.5.1.1, table III
			Disk, type II (chlorine):		
X	X	X	Class 1	4.5.2.2, 4.5.2.6	3.5.1.2, table IV
X	X	X	Class 2	4.5.2.2, 4.5.2.6	3.5.1.2, table IV
X	X	X	Class 3	4.5.2.2, 4.5.2.6	3.5.1.2, table IV
X	X	X	Class 4	4.5.2.2, 4.5.2.6	3.5.1.2, table IV
X	X	X	Class 5	4.5.2.3 4.5.2.6	3.5.1.2, table IV
X	X	X	Class 6	4.5.2.4, 4.5.2.6	3.5.1.2, table IV
X	X	X	Class 7	4.5.2.7	3.5.1.2, table IV
			Thermal shock		
X	X	X	Disks, type I, classes 1, 2, 3, 4, 5, 6, 7, and 8	4.5.2.5	3.4.2
X	X	X	Disks, type II, classes 1, 2, 3, 4, 5, 6, and 7	4.5.2.5	3.4.2

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4.5.2.5 Thermal shock. All colored glass standards shall be removed and immersed for not less than 10 minutes in a container of solid carbon dioxide (dry ice). All surfaces of the colored glass standards shall be in contact with dry ice. At the end of this period, the colored glass standards shall be removed from the dry ice and immediately immersed for not less than 5 minutes in a container of water at a temperature of not less than 200 °F. At the end of this period, delamination, separation, or any other defect of any glass standard shall constitute failure of this test.

4.5.2.6 Comparison. Disks shall be visually compared with the applicable acceptable first article disk in a color comparator (see 3.8). Nonconformance to the applicable acceptable preproduction model disk shall constitute failure of this test.

4.5.2.7 Type II, class 7 disks. For acceptance identical match of all color standards of the offered disks with applicable color standards of a disk which was accepted on a previous contract with the Government, is required. If color identical, testing against chlorine standards may be used as an alternate. Chlorine standards shall be prepared, using the method described for free chlorine samples, in Standard Methods for the Examination of Water and Wastewater. NSN 6810-01-044-0315 Chlorine, Test Tablet, DPD Method shall be used as the indicator. If chlorine solutions are used for matching, tolerances shall be as shown in table III.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of packaging.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105, table I and columns 1 and 2 of table II-A. Inspection level shall be level II.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. The acceptance number shall be zero. No defects shall be allowed.

No.	Defect	A	B	C
111.	Preparation not as specified.	5.1		5.1
112.	Preservatives applied though prohibited.	5.1.1		5.1.1
113.	Cleaning and drying not as specified for level A.	5.2.1.1		
114.	Plastic bag or envelope not as specified for level A.	5.2.1.2		
115.	Filled bags or envelopes not heat sealed as specified for level A.	5.2.1.2		
116.	Unit container not as specified.	5.2.1.3		

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No.	Defect	A	B	C
117.	Technical data, when furnished, not preserved as specified.	5.2.1.3		
118.	Alternate unit container not as specified for level A.	5.2.1.3		
119.	Intermediate container not as specified for level A.	5.2.1.4		5.2.2
120.	Alternate intermediate container not as specified for level A.	5.2.1.4		
121.	Quantity per intermediate container not as specified for level A.	5.2.1.4		
122.	Disks of unlike type and sized placed in same intermediate container.	5.2.1.4		5.2.2
123.	Shipping container not as specified.	5.3.1	5.3.3	5.3.5
124.	Alternate shipping container not as specified.	5.3.2	5.3.4	
125.	Box closure and strapping not as specified.	5.3.1	5.3.3	
126.	Alternate box closure and, as applicable, strapping and water proofing not as specified.	5.3.2	5.3.4	5.3.5
127.	Disks of unlike type and size packed in same shipping container.	5.3.1, 5.3.2	5.3.3, 5.3.4	
128.	Marking not as specified.	5.4.1	5.4.1	5.4.2

5. PACKAGING

5.1 Preparation. Each disk shall be cleaned, thoroughly dried and individually placed in a plastic bag or envelope having a minimum thickness of 2 mils. A dropper (see 3.7) shall be included in the envelope or bag with each type II, class 6 disk. Technical data when furnished, such as an instruction book, an instruction card and a conversion sheet, shall be placed together as a single set in a similar type plastic bag or envelope having a minimum thickness of 2 mils. The plastic bag(s) or envelope(s) shall be sealed closed.

5.1.1 Preservation application. Preservatives shall not be utilized.

5.2 Preservation. Preservation shall be level A or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Cleaning and drying. Cleaning and drying (see 5.1), shall be in accordance with the requirements of MIL-P-116.

5.2.1.2 Plastic bags or envelopes. The plastic bags or envelopes specified in 5.1 shall be manufactured of polyethylene film conforming to L-P-378, type I, class 1. The filled bags or envelopes shall be closed by heat sealing.

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5.2.1.3 Unit container. Each complete disk, prepared and preserved as specified in 5.1 and 5.2.1 respectively, shall be individually placed in a box or container. Furnished technical data, prepared and preserved as specified in 5.1 and 5.2.1 respectively, shall be placed in the box or container with the disk. The unit container or a box conforming to PPP-B-566, variety 1. Alternatively when specified (see 6.2), the unit container shall be a setup box conforming to PPP-B-676, size 8 inches x 4 inches x 1.500 inches. Each container shall be securely closed and, as applicable, the closure shall be in accordance with the appendix to the utilized box specification.

5.2.1.4 Intermediate containers. Twelve complete disks of one type and class, in unit containers as specified in 5.2.1.3, shall be consolidated together in a close-fitting box conforming to PPP-B-636, W6c, style optional. Alternatively, when specified (see 6.2), the intermediate container shall conform to PPP-B-636, type CF, class domestic, grade 150 05 175. Box closure shall be in accordance with the appendix to the box specification.

5.2.2 Level C. Disks, prepared as specified in 5.1, shall be preserved in a manner to afford protection against deterioration and damage from the contractor to the final destination. Only disks of a single type and class shall be consolidated together in a container.

5.3 Packing. Packing shall be level A, level B, or level C as specified (see 6.2).

5.3.1 Level A. Disks of a single type and class, preserved as specified in 5.2, shall be packed in quantities not to exceed 200 pounds in a close-fitting box conforming to PPP-B-601, overseas type, style optional. Box closure and strapping shall be in accordance with the appendix to the box specification except that the strapping shall be flat and zinc coated.

5.3.2 Level A alternate. Alternatively, when specified (see 6.2), two intermediate containers (24 disks) of a single type and class, preserved as specified in 5.2, shall be packed in a close-fitting box conforming to PPP-B-636, class weather resistant. The style and strength of the alternate box shall be based on a type 2 average load, as described in the Appendix to PPP-B-636. Closure, strapping and waterproofing of the alternate box shall be in accordance with the appendix thereto.

5.3.3 Level B. Level B packing shall be as specified in 5.3.1 for level A except that the box shall be domestic type and the strapping may be coated finish.

5.3.4 Level B alternate. Alternate level B packing, when specified (see 6.2), shall be as specified in 5.3.2 for level A alternate except that boxes shall be class domestic. Box closure shall be method I or II in accordance with PPP-B-636.

5.3.5 Level C. Disks of a single type and class, prepared and preserved as specified in 5.1 and 5.2 respectively, shall be packed in a manner to assure carrier acceptance and safe delivery to

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their destination at lowest rates in accordance with Uniform Freight Classification Rates or National Freight Classification Rules.

5.4 Marking. In addition to any special or identification marking required of the contract or purchase order, each container shall be marked in accordance with MIL-STD-129.

6. NOTES

(Thin section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The color disks covered by this specification are intended for use in a comparator conforming to MIL-C-538 to determine pH value and residual chlorine content of water.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. Type and class of disk required (see 1.2).
d. When a first article is required for inspection and approval, and the number of units required (see 3.2).
e. When Government-loaned property is required (see 3.8).
f. Degree of preservation and degree of packing required (see 5.2 and 5.3).
g. When alternate unit container is required (see 5.2.1.3).
h. When alternate level A packing is required (see 5.3.2).
i. When alternate level B packing is required (see 5.3.4).
j. Any special marking (see 5.4.1).

6.3 First article. When a first article inspection is required, the items should be a first production item. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first article. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval in presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

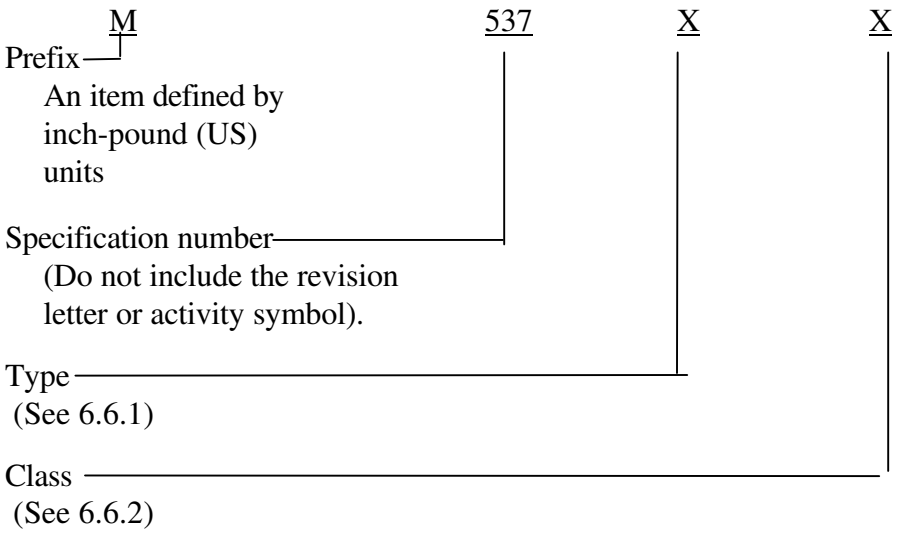
6.4 Government-loaned property. The contracting officer should arrange to loan the property listed in 3.8.

6.5 National Stock Number. This specification is intended to cover disks used in the following item appearing in the Federal Supply Catalog:

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National Stock Number	Item Identification
6630-01-151-6417	Comparator Set, Color, Chlorine-Bromine and pH Determination

6.6 Part or Identifying system. The part or identifying number (PIN) is definitive and formatted to identify each item covered by this specification. The part number in formatted by selecting from the requirement options available in this specification as follows (see 3.6.1):



Examples: M537A1 - Disk, Color Standard, pH, Range 3.0 to 4.6

6.6.1 Type of disk. A one-position alpha field used to designate the type of disk required. Type I disks shall be designated “A”. Type II disks shall be designated “B”.

6.6.2 Class of disk. A one-position numeric field used to designate the class of disk required (see 1.2).

6.7 Subject term (key word) listing.

Color comparator
pH
Residual chlorine

Custodian:
Army - ME
Navy - YD
Air Force - 99

Preparing activity:
Army - ME

Project 6630-0516

Review activity:
Army - MD

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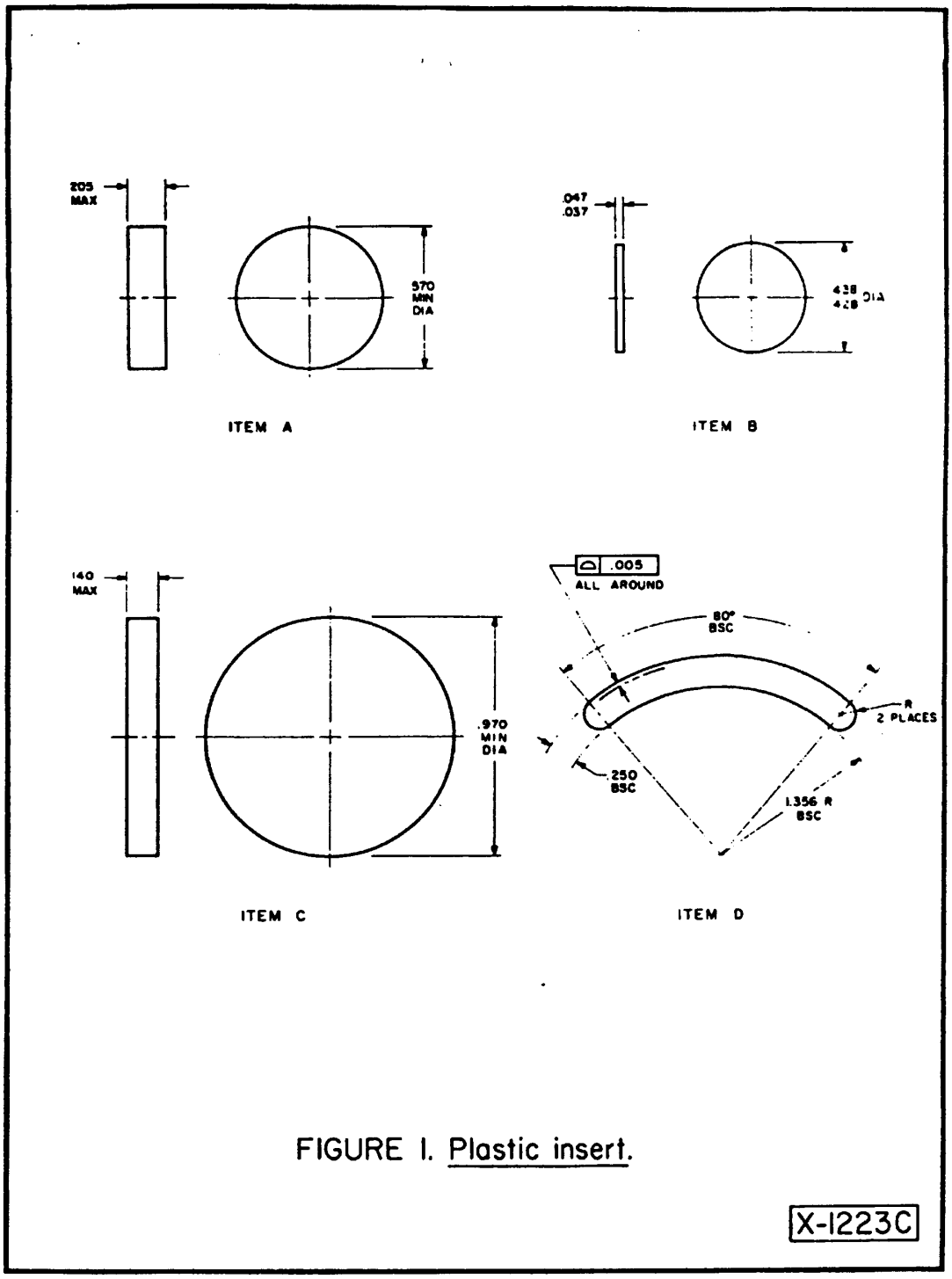


FIGURE 1. Plastic insert.

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